

RIS-Assisted Hybrid Far-Field and Near-Field Source Localization

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ABSTRACT

In this paper, a reconfigurable intelligent surface (RIS)-assisted localization algorithm is proposed for the coexistence of non-line-of-sight (NLOS) far-field (FF) and near-field (NF) users. To estimate the angle and distance parameters of both FF and NF users, the mixed signals at the RIS are first recovered using the least squares method. Subsequently, a conventional two-dimensional (2D) MUSIC algorithm is applied to obtain all candidate angles of mixed signals. Furthermore, a spatial differencing method is employed to separate the NF components. A reduced-dimension MUSIC method is then applied to the extracted three-dimensional (3D) parameters of NF signals. Finally, by integrating the two-stage MUSIC-based parameter estimators, the classification of FF and NF users is achieved. Simulations demonstrate the effectiveness of the proposed algorithm.

Index Terms— RIS, Near-field, non-line-of-sight, spatial differencing.

1. INTRODUCTION

With ever increasing demands of wireless communication systems, reconfigurable intelligent surface (RIS) has become a key enabling technology for 6G communications [1–5]. RIS consists of a large number of controllable passive elements whose electromagnetic parameters can be tuned to precisely modulate incident signals [4], thereby reshaping the wireless propagation environment. When the line-of-sight (LOS) path between the base station (BS) and user equipment (UE) is blocked, RIS can support communication by generating virtual NLOS reflection paths.

In recent years, RIS-assisted localization has attracted growing attention; however, existing studies have primarily focused on either FF source localization [6–8] or NF source localization [9–11]. However, in many practical applications, FF and NF signals often coexist, and models designed for a

single scenario are difficult to apply directly to such hybrid-source environments. Several existing studies have proposed various hybrid FF and NF source localization methods for conventional antenna arrays [12–15]; however, research in RIS-assisted scenarios remains rather limited.

This paper proposes a RIS-assisted hybrid FF and NF source localization algorithm for distinguishing and localizing multiple FF and NF users. First, the incident signals at the RIS are recovered using the least squares (LS) method to obtain complete observation signals. Subsequently, a conventional two-dimensional MUSIC algorithm is employed to obtain the angular information of the mixed signals, and the mixed components are separated by exploiting the distinct covariance matrix properties of FF and NF signals. Then, based on the separated pure NF signals, a reduced-dimension MUSIC method is applied, in which the 3D parameter estimation is decomposed into three one-dimensional estimations, significantly reducing the computational complexity. Finally, the NF angles are removed from the set of angular estimates to obtain the FF angle set. Simulation results demonstrate that the proposed algorithm maintains high accuracy and stability under different SNR levels and snapshot conditions.

2. SIGNAL MODEL

This paper considers an RIS-assisted localization system, illustrated in Fig. 1, where the BS is equipped with M_B antennas, and the RIS is composed of N_R elements arranged in a uniform planar array (UPA). The RIS is placed on the YOZ plane, where each row contains N_y elements and each column contains N_z elements. Thus, the total number of RIS elements is $N_R = N_y N_z$. Assume N_y and N_z are odd numbers, and the central element of the RIS is selected as the reference element, whose coordinates are $\mathbf{p}_R = [0, 0, 0]^T$. K UEs are to be localized, with K_1 UEs in the NF region and K_2 in the FF region, satisfying $K = K_1 + K_2$. Since the LOS path between the UE and the BS is blocked, localization relies on the reflected path from the UE to the RIS and then to the BS. Because the positions of the BS and the RIS are fixed, the channel between them is assumed to be known. Let d_y and d_z denote the element spacing of the RIS along the row and column directions, respectively, and assume $d_y = d_z = d$ with $d \leq \lambda/4$, where λ represents

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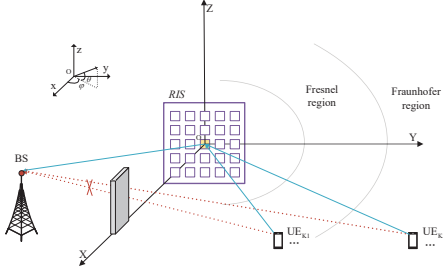


Fig. 1. Geometric model.

the signal wavelength. In addition, define p and q as the row and column indices of the RIS elements, respectively, where $p \in \{-\frac{N_y-1}{2}, \dots, \frac{N_y-1}{2}\}$, $q \in \{-\frac{N_z-1}{2}, \dots, \frac{N_z-1}{2}\}$. The array aperture is $D = \max\{(N_y - 1)d, (N_z - 1)d\}$, and the Fresnel region is $\left[0.62\sqrt{\frac{D^3}{\lambda}}, \frac{2D^2}{\lambda}\right]$.

The k -th UE is located at $r_k[\cos \varphi_k \cos \theta_k, \sin \varphi_k \cos \theta_k, \sin \theta_k]$ where r_k denotes the distance from the UE to the reference element of the RIS, while (φ_k, θ_k) represent the azimuth and elevation angles from the UE to the RIS, respectively. The (p, q) -th RIS element has coordinates $\mathbf{p}_{(p,q)} = [0, pd, qd]^T$, and the distance between the k -th UE and the (p, q) -th RIS element, denoted by $r_k^{(p,q)}$, is given by

$$\begin{aligned} r_k^{(p,q)} &= \sqrt{r_k^2 + p^2 d^2 + q^2 d^2 - 2r_k p d \cos \theta_k \sin \varphi_k - 2r_k q d \cos \varphi_k} \\ &\approx r_k - p d \sin \varphi_k \cos \theta_k - q d \sin \theta_k + \frac{p^2 d^2 (1 - \sin^2 \varphi_k \cos^2 \theta_k)}{2r_k} \\ &\quad + \frac{q^2 d^2 \cos^2 \theta_k}{2r_k}. \end{aligned} \quad (1)$$

In (1), the final step of the derivation is obtained by applying the Fresnel approximation and ignoring second-order terms. In the NF region, the steering vector of the k -th user toward the RIS is denoted by $\mathbf{b}(\theta_k, \varphi_k, r_k) \in \mathbb{C}^{N_R \times 1}$. Then, for the (p, q) -th element of the RIS, the response is given by

$$[\mathbf{b}(\theta_k, \varphi_k, r_k)]_{(p,q)} = \exp \left[-j \frac{2\pi}{\lambda} (r_k^{(p,q)} - r_k) \right]. \quad (2)$$

Substituting (1) into (2) yields,

$$[\mathbf{b}(\theta_k, \varphi_k, r_k)]_{(p,q)} \approx \exp \left[-j \frac{2\pi}{\lambda} (\alpha_k + \beta_k) \right]. \quad (3)$$

Therefore, the phase of the steering vector for an N-F signal is decomposed into two components, namely $\alpha_k = -pd \sin \varphi_k \cos \theta_k + \frac{p^2 d^2 (1 - \sin^2 \varphi_k \cos^2 \theta_k)}{2r_k}$ and $\beta_k = -qd \sin \theta_k + \frac{q^2 d^2 \cos^2 \theta_k}{2r_k}$. By decomposing the steering vector along the Y and Z directions, we obtain the following results

$$\mathbf{b}(\theta_k, \varphi_k, r_k) = \mathbf{a}_Z(\theta_k, r_k) \otimes \mathbf{a}_Y(\theta_k, \varphi_k, r_k), \quad (4)$$

$$\mathbf{a}_Y(\theta_k, \varphi_k, r_k) = \exp \left(-j \frac{2\pi}{\lambda} \alpha_k \right), \quad (5)$$

$$\mathbf{a}_Z(\theta_k, r_k) = \exp \left(-j \frac{2\pi}{\lambda} \beta_k \right). \quad (6)$$

In the FF region, the distance r_k from the UE to the reference element of the RIS is much larger than the distance from any other RIS element to the reference element, i.e., $(pd)^2 + (qd)^2 \ll r_k^2$. At this point, the second-order curvature term can be ignored and only the first-order linear phase term is retained. The FF steering vector of the k -th UE is

$$\mathbf{b}(\theta_k, \varphi_k) = \mathbf{a}_Z(\theta_k) \otimes \mathbf{a}_Y(\theta_k, \varphi_k), \quad (7)$$

$$\mathbf{a}_Y(\theta_k, \varphi_k) = e^{-j \frac{2\pi}{\lambda} (-pd \sin \varphi_k \cos \theta_k)}, \quad (8)$$

$$\mathbf{a}_Z(\theta_k) = e^{-j \frac{2\pi}{\lambda} (-qd \sin \theta_k)}. \quad (9)$$

In time-slot t , the signal $s_k(t)$ transmitted by the k -th UE is a zero-mean random signal satisfying $\mathbb{E}\{s_k(t)s_k^*(t)\} = 1$. In addition, all K UE signals are assumed to be mutually independent. The signal received at the BS can be expressed as

$$\mathbf{y}(t) = \sum_{k=1}^K \mathbf{H}_{RB} \mathbf{\Psi} \mathbf{g}_{UR,k} s_k(t) + \mathbf{w}(t), \quad (10)$$

where $\mathbf{H}_{RB} \in \mathbb{C}^{M_{Bs} \times N_R}$ denotes the known channel between the RIS and the BS, and $\mathbf{\Psi} = \text{diag}\{e^{j\phi_1}, \dots, e^{j\phi_{N_R}}\}$ represents the reflection matrix of the RIS. The vector $\mathbf{g}_{UR,k} = \sqrt{\eta_k} \mathbf{b}(\theta_k, \varphi_k, r_k) \in \mathbb{C}^{N_R \times 1}$ or $\mathbf{g}_{UR,k} = \sqrt{\eta_k} \mathbf{b}(\theta_k, \varphi_k) \in \mathbb{C}^{N_R \times 1}$ denotes the NF or FF channel between the k -th UE and the RIS, where η_k is the path-loss coefficient. In addition, $\mathbf{w}(t) = [w_1(t), \dots, w_{M_{Bs}}(t)]^T$ denotes the additive independent complex Gaussian noise with variance σ^2 across all BS antennas.

For the hybrid signal model, (10) can be compactly expressed as:

$$\mathbf{y}(t) = \mathbf{H}_{RB} \mathbf{\Psi} [\mathbf{A}_N \mathbf{s}_N(t) + \mathbf{A}_F \mathbf{s}_F(t)] + \mathbf{w}(t), \quad (11)$$

where $\mathbf{s}_N(t) = [\sqrt{\eta_1} s_1(t), \sqrt{\eta_2} s_2(t), \dots, \sqrt{\eta_{K_1}} s_{K_1}(t)]^T$, $\mathbf{s}_F(t) = [\sqrt{\eta_1} s_1(t), \sqrt{\eta_2} s_2(t), \dots, \sqrt{\eta_{K_2}} s_{K_2}(t)]^T$, $\mathbf{A}_N = [\mathbf{b}(\theta_1, \varphi_1, r_1), \mathbf{b}(\theta_2, \varphi_2, r_2), \dots, \mathbf{b}(\theta_{K_1}, \varphi_{K_1}, r_{K_1})]$, and $\mathbf{A}_F = [\mathbf{b}(\theta_1, \varphi_1), \mathbf{b}(\theta_2, \varphi_2), \dots, \mathbf{b}(\theta_{K_2}, \varphi_{K_2})]$.

3. THE PROPOSED ALGORITHM

The DOA and distance of the K UEs are contained in the steering matrix $\mathbf{A}$. To extract them, the components $\mathbf{A}_N \mathbf{s}_N(t) + \mathbf{A}_F \mathbf{s}_F(t)$ must be recovered from the received signal $\mathbf{y}(t)$. Let $\mathbf{\Gamma} = \mathbf{H}_{RB} \mathbf{\Psi}$ and define $\mathbf{A}_N \mathbf{s}_N(t) + \mathbf{A}_F \mathbf{s}_F(t)$; we then obtain the LS estimate $\bar{\mathbf{x}}(t) = \mathbf{\Gamma}^\dagger \mathbf{y}(t)$, where $\mathbf{\Gamma}^\dagger$ is the pseudoinverse of $\mathbf{\Gamma}$. For the LS estimate to be unique, $\mathbf{\Gamma}$ must have full column rank. Since $M_{Bs} < N_R$, there exists $\text{rank}(\mathbf{\Gamma}) \leq \min(M_{Bs}, N_R) = M_{Bs}$. In this

case, since the number of unknowns exceeds the number of observations, the LS estimate of $\bar{\mathbf{x}}(t)$ cannot yield a correct solution. To address this issue, the time interval can be divided into N_L sub-slots, and the RIS phase can be changed in each sub-slot. In each time-slot, the RIS adopts N_L different configurations. The effective matrix $\tilde{\mathbf{\Gamma}}$ is $\tilde{\mathbf{\Gamma}} = (\mathbf{I}_{N_L} \otimes \mathbf{H}_{RB})\tilde{\mathbf{\Psi}}$, where $\tilde{\mathbf{\Psi}} = [\tilde{\mathbf{\Psi}}_1, \dots, \tilde{\mathbf{\Psi}}_{N_L}]^T$. Accordingly, the received signal at the BS at time-slot t becomes

$$\tilde{\mathbf{y}}(t) = [\mathbf{y}^{(1)}(t) \quad \mathbf{y}^{(2)}(t) \quad \dots \quad \mathbf{y}^{(N_L)}(t)]^T. \quad (12)$$

When N_L is sufficiently large, $\tilde{\mathbf{\Gamma}}$ becomes full column rank, and the unique estimate of $\mathbf{x}(t)$ is

$$\bar{\mathbf{x}}(t) = \tilde{\mathbf{\Gamma}}^\dagger \tilde{\mathbf{y}}(t) = (\tilde{\mathbf{\Gamma}}^H \tilde{\mathbf{\Gamma}})^{-1} \tilde{\mathbf{\Gamma}}^H \tilde{\mathbf{y}}(t). \quad (13)$$

Therefore, under ideal conditions, the covariance matrix can be written as

$$\begin{aligned} \mathbf{R} &= \frac{1}{T} \sum_{t=1}^T \bar{\mathbf{x}}(t) (\bar{\mathbf{x}}(t))^H \\ &= \mathbf{R}_N + \mathbf{R}_F + \sigma^2 \mathbf{I} \\ &= \mathbf{A}_N \mathbf{S}_N \mathbf{A}_N^H + \mathbf{A}_F \mathbf{S}_F \mathbf{A}_F^H + \sigma^2 \mathbf{I}_{N_R}, \end{aligned} \quad (14)$$

where $\mathbf{R}_N$ and $\mathbf{R}_F$ are the covariance matrices of the NF and FF components, $\mathbf{S}_N$ and $\mathbf{S}_F$ are the corresponding source covariance matrices, and T is the number of snapshots. Here, $(\cdot)^H$ denotes the Hermitian transpose. Then, the eigenvalue decomposition (EVD) of $\mathbf{R}$ is done to achieve the noise subspace matrix $\mathbf{U}_N \in \mathbb{C}^{N_R \times (N_R - K)}$.

To obtain all angular information from the hybrid signals, this paper employs the 2D MUSIC algorithm by searching over the elevation θ_k and azimuth φ_k as follows

$$P_{\text{MUSIC}}(\theta, \varphi) = \frac{1}{\mathbf{b}^H(\theta, \varphi) \mathbf{U}_N \mathbf{U}_N^H \mathbf{b}(\theta, \varphi)}. \quad (15)$$

Based on the Toeplitz property of FF covariance matrix, we have

$$\mathbf{R}_F^T = \mathbf{F} \mathbf{R}_F \mathbf{F}, \quad (16)$$

where $\mathbf{F} \in \mathbb{R}^{N_R \times N_R}$ is a permutation matrix whose anti-diagonal entries are all equal to 1 and all other entries are 0. According to (16), the FF component can be separated to obtain the pure NF component as

$$\Delta \mathbf{R} = \mathbf{F} \mathbf{R} \mathbf{F} - \mathbf{R}^T = \mathbf{F} \mathbf{R}_N \mathbf{F} - \mathbf{R}_N^T. \quad (17)$$

Unlike the oblique projection technique, the matrix $\Delta \mathbf{R}$ contains only NF components. Since $\mathbf{R}_N^T = \mathbf{R}_N^*$ and $\mathbf{F} \mathbf{F} = \mathbf{I}$, the covariance matrix can be further expressed as

$$\begin{aligned} \mathbf{R}_D &= \mathbf{F} \Delta \mathbf{R} \mathbf{F} \\ &= \mathbf{R}_N - \mathbf{F} \mathbf{R}_N^* \mathbf{F} \\ &= [\mathbf{A}_N \quad \mathbf{F} \mathbf{A}_N^*] \begin{bmatrix} \mathbf{S}_N & \\ & -\mathbf{S}_N \end{bmatrix} [\mathbf{A}_N \quad \mathbf{F} \mathbf{A}_N^*]^H \\ &= \mathbf{A}_D \mathbf{S}_D \mathbf{A}_D^H. \end{aligned} \quad (18)$$

The EVD of $\mathbf{R}_D$ can obtain its noise subspace $\mathbf{U}_n \in \mathbb{C}^{N_R \times (N_R - 2K_1)}$. Then, a reduced-dimension MUSIC approach is adopted, which decomposes the 3D estimation into three one-dimensional estimations. Based on the symmetry of the array structure, (6) can be decomposed into

$$\begin{aligned} \mathbf{a}_Z(\theta_k, r_k) &= \exp\left(-j \frac{2\pi}{\lambda} \beta_k\right) \\ &= \begin{bmatrix} e^{j(-q)\omega_z} & & & \\ & e^{j(-q+1)\omega_z} & & \\ & & \ddots & \\ & & & 1 \\ & & & & \ddots \\ & & & & & 1 \\ & & & & & & e^{j(q-1)\omega_z} \\ & & & & & & & e^{jq\omega_z} \end{bmatrix} \begin{bmatrix} e^{j(-q)^2\phi_z} \\ e^{j(-q+1)^2\phi_z} \\ \vdots \\ e^{j(-1)^2\phi_z} \\ 1 \end{bmatrix} \\ &= \zeta_Z(\theta_k) \mathbf{v}_Z(\theta_k, r_k), \end{aligned} \quad (19)$$

where $\omega_z = -\frac{2\pi}{\lambda} d \sin \theta_z$ and $\phi_z = -\frac{2\pi}{\lambda} \frac{d^2 \cos^2 \theta_k}{2r_k}$. Then, (4) can be written as

$$\mathbf{b}(\theta_k, \varphi_k, r_k) = (\zeta_Z(\theta_k) \mathbf{v}_Z(\theta_k, r_k)) \otimes (\mathbf{I}_Y \mathbf{a}_Y(\theta_k, \varphi_k, r_k)). \quad (20)$$

Similarly, we obtain

$$\begin{aligned} \mathbf{a}_Y(\theta_k, \varphi_k, r_k) &= \exp\left(-j \frac{2\pi}{\lambda} \alpha_k\right) \\ &= \begin{bmatrix} e^{j(-p)\omega_y} & & & \\ & e^{j(-p+1)\omega_y} & & \\ & & \ddots & \\ & & & 1 \\ & & & & \ddots \\ & & & & & 1 \\ & & & & & & e^{j(p-1)\omega_y} \\ & & & & & & & e^{jp\omega_y} \end{bmatrix} \begin{bmatrix} e^{j(-p)^2\phi_y} \\ e^{j(-p+1)^2\phi_y} \\ \vdots \\ e^{j(-1)^2\phi_y} \\ 1 \end{bmatrix} \\ &= \zeta_Y(\varphi_k) \mathbf{v}_Y(\varphi_k, r_k), \end{aligned} \quad (21)$$

where $\omega_y = -\frac{2\pi}{\lambda} d \sin \varphi_k \cos \theta_k$ and $\phi_y = -\frac{\pi d^2}{\lambda r_k} (1 - \sin^2 \varphi_k \cos^2 \theta_k)$.

Furthermore, based on the properties of the Kronecker product $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$, $\mathbf{A}_D = [\mathbf{A}_N \quad \mathbf{F} \mathbf{A}_N^*]$ can be further written as

$$\begin{aligned} \mathbf{A}_D &= [\mathbf{A}_N \quad \mathbf{F} \mathbf{A}_N^*] \\ &= \mathbf{C}(\theta_k) [(\mathbf{v}_Z(\theta_k, r_k) \otimes \mathbf{a}_Y(\theta_k, \varphi_k, r_k)) \\ &\quad (\mathbf{v}_Z^*(\theta_k, r_k) \otimes \mathbf{F}_2 \mathbf{a}_Z^*(\theta_k, \varphi_k, r_k))], \end{aligned} \quad (22)$$

where $\mathbf{C}(\theta_k) = (\zeta_Z(\theta_k) \otimes \mathbf{I}_Y)$, $\mathbf{F} = \mathbf{F}_1 \otimes \mathbf{F}_2$ with $\mathbf{F}_1 \in \mathbb{R}^{N_z \times N_z}$ and $\mathbf{F}_2 \in \mathbb{R}^{N_y \times N_y}$ represent the permutation matrices in the Z and Y directions, respectively.

Accordingly, the first-stage search can be performed to estimate the elevation of the NF signals as follows

$$\hat{\theta} = \arg \max_{\theta} \frac{1}{\det[\mathbf{D}(\theta_k)]}, \quad (23)$$

where $\mathbf{D}(\theta_k) = \mathbf{C}^H(\theta_k) \mathbf{U}_n \mathbf{U}_n^H \mathbf{C}(\theta_k)$.

Similarly, $\mathbf{A}_D$ can be further decomposed as

$$\begin{aligned}\mathbf{A}_D &= [\mathbf{A}_N \quad \mathbf{F}\mathbf{A}_N^*] \\ &= \mathbf{C}(\theta_k)\mathbf{E}(\varphi_k)[(\mathbf{v}_Z(\theta_k, r_k) \otimes \mathbf{v}_Y(\theta_k, \varphi_k, r_k)) \\ &\quad (\mathbf{v}_Z^*(\theta_k, r_k) \otimes \mathbf{v}_Y^*(\theta_k, \varphi_k, r_k))],\end{aligned}\quad (24)$$

where $\mathbf{E}(\varphi_k) = (\mathbf{I}_{Z+1} \otimes \boldsymbol{\zeta}_Y(\varphi_k))$. Based on (24), a second MUSIC spectrum can be constructed to estimate the azimuth of the NF signals as follows

$$\hat{\varphi} = \arg \max_{\varphi} \frac{1}{\det[\mathbf{F}(\hat{\theta}, \varphi_k)]}. \quad (25)$$

where $\mathbf{F}(\hat{\theta}, \varphi_k) = \mathbf{E}^H(\varphi_k)\mathbf{C}^H(\hat{\theta})\mathbf{U}_n\mathbf{U}_n^H\mathbf{C}(\hat{\theta})\mathbf{E}(\varphi_k)$.

Substituting $\hat{\theta}$ and $\hat{\varphi}$ into the NF steering vector $\mathbf{b}(\theta_k, \varphi_k, r_k)$ yields

$$\mathbf{V}(\hat{\theta}, \hat{\varphi}, r_k) = \mathbf{b}^H(\hat{\theta}, \hat{\varphi}, r_k)\mathbf{U}_n\mathbf{U}_n^H\mathbf{b}(\hat{\theta}, \hat{\varphi}, r_k). \quad (26)$$

The NF distance r_k is estimated by

$$\hat{r}_k = \arg \max_{r_k} \frac{1}{\mathbf{V}(\hat{\theta}, \hat{\varphi}, r_k)}. \quad (27)$$

After obtaining the NF parameters, the FF angles can be extracted by removing the NF angles from all candidate angles.

4. SIMULATION RESULTS

In this section, the number of users is set to $K = 4$, including $K_1 = 2$ NF users located at $(30^\circ, -60^\circ, 0.5m)$ and $(-30^\circ, 60^\circ, 1.0m)$ and $K_2 = 2$ FF users located at $(50, -40)$ and $(-50, 40)$. Unless otherwise specified, the RIS is configured as a UPA composed of $N_R = 25 \times 25$ reflecting elements. The BS employs a uniform linear array with $M_B = 128$ antennas, and the element spacing is set to a quarter wavelength. The required number of sub-slots is $N_L = \lceil \frac{N_R}{M_B} \rceil$. The number of snapshots is $T = 300$, and the number of Monte Carlo trials is 500. In addition, the RIS-BS channel is assumed to be perfectly known, and it follows a Rician fading model with a Rician factor of 2. Since the LOS path is completely blocked, the localization relies solely on the reflected UE-RIS-BS link.

Fig. 2(a) shows the 2D MUSIC spectrum at an SNR of 10 dB. Sharp peaks can be observed at the true user angles, corresponding to both the FF and NF users. This demonstrates that the 2D MUSIC algorithm can provide a complete set of angle candidates for mixed signals. Fig. 2(b) illustrates the joint estimation results of the angle and distance for the NF signals. The estimated points are close to the true positions, which confirms that the proposed reduced-dimension MUSIC algorithm achieves good estimation accuracy.

To evaluate the performance of the proposed algorithm under different SNR conditions, the RMSE curves are plotted for SNR values ranging from -10 dB to 10 dB, as shown in

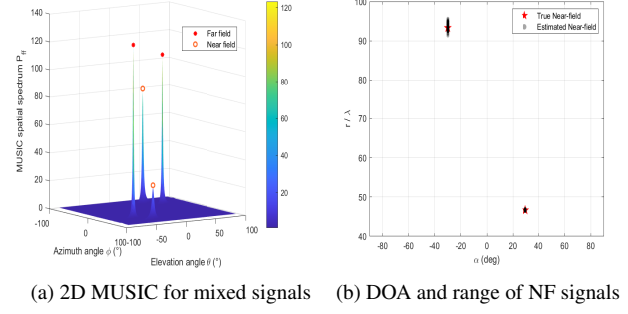


Fig. 2. Parameters estimation in mixed-source scenarios.

Fig. 3(a)-(b). Compared with the Parisa algorithm [11], the proposed method significantly reduces the RMSE in angle estimation. In terms of distance estimation, the two algorithms show similar trends, but the proposed method performs slightly better in high SNR regions. Fig. 3(c) presents the RMSE variation of the FF signals. As the SNR increases, the error decreases rapidly and becomes stable in the medium- and high-SNR regions.

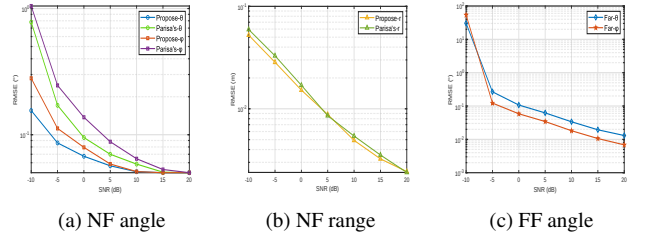


Fig. 3. RMSE versus SNR.

5. CONCLUSIONS

In this paper, a RIS-assisted hybrid FF and NF localization method has been proposed. The received signal is first retrieved using LS, and sub-slot division with phase variations forms a full-rank observation matrix. The 2D MUSIC algorithm is used to estimate the angles of mixed signals. Using the property differences of covariance matrices, the spatial-differencing technique separates the FF and NF components. The NF's DOA and range parameters are then obtained with the reduced-dimension MUSIC algorithm, and the FF's DOA parameters are extracted by removing the NF information. Simulations verify the correctness of the proposed method.

6. REFERENCES

- [1] Q. Wu, S. Zhang, B. Zheng, C. You, and R. Zhang, "Intelligent reflecting surface-aided wireless communications: A tutorial," *IEEE Transactions on Communications*, vol. 69, no. 5, pp. 3313–3351, 2021.

- [2] X. Yuan, Y.-J. A. Zhang, Y. Shi, W. Yan, and H. Liu, "Reconfigurable-intelligent-surface empowered wireless communications: Challenges and opportunities," *IEEE Wireless Communications*, vol. 28, no. 2, pp. 136–143, 2021.
- [3] M. A. ElMossallamy, H. Zhang, L. Song, K. G. Seddik, Z. Han, and G. Y. Li, "Reconfigurable intelligent surfaces for wireless communications: Principles, challenges, and opportunities," *IEEE Transactions on Cognitive Communications and Networking*, vol. 6, no. 3, pp. 990–1002, 2020.
- [4] Y. Liu, X. Liu, X. Mu, T. Hou, J. Xu, M. Di Renzo, and N. Al-Dhahir, "Reconfigurable intelligent surfaces: Principles and opportunities," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 3, pp. 1546–1577, 2021.
- [5] E. C. Strinati, G. C. Alexandropoulos, H. Wymeersch, B. Denis, V. Sciancalepore, R. D'Errico, A. Clemente, D.-T. Phan-Huy, E. De Carvalho, and P. Popovski, "Reconfigurable, intelligent, and sustainable wireless environments for 6g smart connectivity," *IEEE Communications Magazine*, vol. 59, no. 10, pp. 99–105, 2021.
- [6] J. He, A. Fakhreddine, C. Vanwynsberghe, H. Wymeersch, and G. C. Alexandropoulos, "3d localization with a single partially-connected receiving ris: Positioning error analysis and algorithmic design," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 10, pp. 13 190–13 202, 2023.
- [7] D.-R. Emenonye, H. S. Dhillon, and R. M. Buehrer, "Fundamentals of ris-aided localization in the far-field," *IEEE Transactions on Wireless Communications*, vol. 23, no. 4, pp. 3408–3424, 2024.
- [8] K. Keykhosravi, M. F. Keskin, G. Seco-Granados, and H. Wymeersch, "Siso ris-enabled joint 3d downlink localization and synchronization," in *ICC 2021 - IEEE International Conference on Communications*, 2021, pp. 1–6.
- [9] H. Yan, H. Chen, W. Liu, S. Yang, G. Wang, and C. Yuen, "Ris-enabled joint near-field 3d localization and synchronization in siso multipath environments," *IEEE Transactions on Green Communications and Networking*, vol. 9, no. 1, pp. 367–379, 2025.
- [10] Z. Abu-Shaban, K. Keykhosravi, M. F. Keskin, G. C. Alexandropoulos, G. Seco-Granados, and H. Wymeersch, "Near-field localization with a reconfigurable intelligent surface acting as lens," in *ICC 2021 - IEEE International Conference on Communications*, 2021, pp. 1–6.
- [11] P. Ramezani, A. Kosasih, and E. Bjornson, "An efficient modified music algorithm for ris-assisted near-field localization," in *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, 2024, pp. 4430–4435.
- [12] J. Liang and D. Liu, "Passive localization of mixed near-field and far-field sources using two-stage music algorithm," *IEEE Transactions on Signal Processing*, vol. 58, no. 1, pp. 108–120, 2010.
- [13] J. He, M. N. S. Swamy, and M. O. Ahmad, "Efficient application of music algorithm under the coexistence of far-field and near-field sources," *IEEE Transactions on Signal Processing*, vol. 60, no. 4, pp. 2066–2070, 2012.
- [14] G. Liu and X. Sun, "Two-stage matrix differencing algorithm for mixed far-field and near-field sources classification and localization," *IEEE Sensors Journal*, vol. 14, no. 6, pp. 1957–1965, 2014.
- [15] Z. Zheng, M. Fu, D. Jiang, W.-Q. Wang, and S. Zhang, "Localization of mixed far-field and near-field sources via cumulant matrix reconstruction," *IEEE Sensors Journal*, vol. 18, no. 18, pp. 7671–7680, 2018.